Context-Aligned Large Language Models for Situational Awareness in Critical Grid Events

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***Abstract*—Power system operators require real-time situational awareness to maintain grid reliability during normal and emergency conditions. During grid emergencies, the operator decision-making process is always high-stakes. Insufficient situational awareness, such as incorrect or untimely responses, can be the cause for major blackouts resulting in adverse impacts on millions of customers as well as huge economic losses. Traditional emergency management systems lack the ability to synthesize multi-source regulatory and operational information during time-critical scenarios. This paper presents a Retrieval-Augmented Generation (RAG) system for power system operator decision support leveraging a Large Language Model (LLM) with domain-specific knowledge retrieval to address three operational queries: emergency procedure lookup, compliance interpretation, and event response. This involves hybrid retrieval combining sparse and dense searches over local documents, followed by prompting a local quantized LLM. Two critical features of the proposed RAG-LLM framework are the ability to preserve regulatory metadata and achieve rapid query processing on commodity GPU hardware for supporting operators during grid emergencies. Experimental results indicate that the proposed framework has a higher degree of potential to enhance power system operator situational awareness in critical scenarios.**

***Index Terms*—GPU, large language model (LLM), power system emergency operations, retrieval-augmented generation (RAG), situational awareness (SA).**

I. Introduction

Modern power system operations require continuous coordination between generation, transmission, and load to maintain grid stability and reliability. Power systems commonly operate across multiple operation states from normal economic dispatch to emergency conditions requiring precise restorative actions [1]. Under critical conditions, power system operators are tasked with managing increasingly complex decision-making scenarios as system stress escalates. Insufficient situational awareness (SA) has been identified as a primary cause for major blackouts. For instance, the August 14, 2003 blackout that affected approximately 50 million people and caused $4–$10 billion in economic losses across the U.S. and Canada demonstrated the consequences of inadequate SA and thorough understanding of the system [2].

Subsequent research has reinforced this finding, with comprehensive studies identifying a lack of SA during grid emergencies as directly leading to delayed or incorrect operator responses with catastrophic consequences [3]. Amplifying these challenges, traditional alarm-management systems in modern control centers overwhelm the power system operators with information floods during critical events. For instance, in the context of cognitive load in energy network control, Afzal *et al.* [4] reported that over 50% of alarms during emergencies are redundant, decreasing SA rather than enhancing it. These challenges highlight the growing need for advanced decision-support systems that can intelligently organize and contextualize procedural knowledge to efficiently support operator SA as well as decision-making.

Recent advances in AI have demonstrated high capability in processing and synthesizing complex information. Most prominently, $Large\ Language\ Models\ (LLMs)$ are increasingly receiving attention as these models leverage self-attention mechanisms to capture long-range dependencies and contextualize semantic relationships, enabling sophisticated parametric reasoning. In this regard, the transformer architecture proposed by Vaswani *et al.* [5] is widely adopted to serve as intelligent assistants across diverse domains, from medical diagnosis to scientific discovery [6]. However, despite their impressive capabilities, pre-trained LLMs face fundamental limitations when applied to specialized domains; they lack access to up-to-date, domain-specific documentation critical for grounded answers to operator queries. Moreover, the lack of said knowledge makes these models prone to hallucination, i.e., generating plausible but factually incorrect information. To address these limitations, Lewis *et al.* [7] introduced $Retrieval\text{-}Augmented\ Generation\ (RAG)$ as a paradigm that combines information retrieval with sequence-to-sequence generation, enabling models to access non-parametric knowledge without computationally expensive and redundant retraining. RAG systems augment LLM prompts with relevant document passages retrieved from external knowledge bases, grounding model output in verifiable sources while maintaining the flexibility of pre-trained language models. Chen *et al.* [8] demonstrated that this architecture is particularly effective for knowledge-intensive tasks requiring factual accuracy and citation traceability, making it well suited for high-stakes applications such as power system operations.

Recent studies have explored LLM and RAG frameworks for power system operations, including multi-agent systems for autonomous grid control [9], benchmark frameworks evaluating LLMs on dispatch optimization tasks [10], and grid-aware multi-modal RAG systems for fault diagnosis [11]. Domain-specific

This work is based in part upon work supported by the National Science Foundation (NSF) through the ASPIRE Engineering Research Center under Award# 1941524, NSF–Office of Advanced Cyberinfrastructure (OAC) under Award# 2346717, and the Department of Energy/ National Nuclear Security Administration under Award# DE-NA0004016.

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RAG frameworks for electric power system query-answering have demonstrated improved accuracy through hybrid retrieval strategies that employ sparse and dense methods, alongside adaptive document segmentation methods [12]. However, no prior work has explicitly integrated North American Electric Reliability Corporation (NERC) reliability standards with regional Independent System Operator (ISO) operating procedures while addressing three operational query types: emergency procedure lookup, compliance interpretation, and event response guidance that collectively support the three dimensions of SA, i.e., (i) *perception* of system status, (ii) *comprehension* of operational significance, and (iii) *projection* of future grid conditions [13], [14].

This paper presents a hybrid RAG-LLM framework specifically designed to enhance SA in power system operations during grid emergencies through domain-adapted knowledge retrieval and contextualized response generation. Our approach integrates NERC reliability standards, regional ISO operating procedures, and historical incident reports into a unified corpus with preserved metadata. The system combines Best Matching 25 (BM25) sparse retrieval, Facebook AI Similarity Search (FAISS) dense vector search, and Reciprocal Rank Fusion (RRF) to optimize hybrid retrieval performance across diverse queries. A 4-bit quantized Llama 3.1-8B-Instruct model processes the retrieved contexts to generate evidence-grounded responses with full source traceability, ensuring factual accuracy in high-stakes operational decisions. The major contributions and advantage of this paper over existing approaches lie in the following: **(1)** Development of a hybrid retrieval system that combines BM25 sparse matching, FAISS dense search, and RRF ranking optimized for power system regulatory and operational documentation; **(2)** A comprehensive analysis of a curated corpus of NERC reliability standards, regional ISO procedures, and incident reports, paired with complementary metadata to support three major operational query types: compliance interpretation, procedure lookup, and event response; **(3)** Development of an effective RAG-LLM framework explicitly mapping the types to the three dimensions of SA, consisting of perception, comprehension, and projection; **(4)** Development of efficient context-aligned algorithms enabling rapid inference on commodity GPU hardware for real-time decision support, thereby contributing to the field of machine learning; and **(5)** Advancement in the field of machine learning (via LLMs) by providing a critical foundation toward grid modernization. This paper thus advances the application of LLMs and RAG techniques to enhance the SA and decision support in power system operations.

This paper is organized as follows. Section II presents the proposed system architecture and hybrid retrieval methodology. Section III and Section IV present the experimental setup including evaluation metrics, and results and discussion, respectively. Key findings of the paper including future directions are presented in Section V.

II. Proposed Integrated RAG-LLM Framework

This section details the proposed RAG-LLM framework, which is conceptually illustrated in Fig. 1, with a goal to

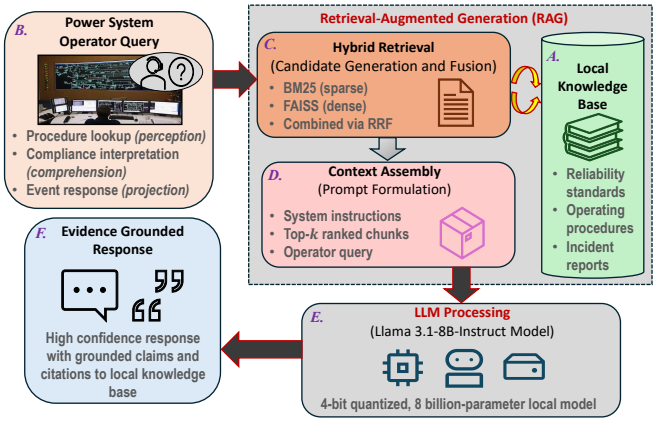

Fig. 1. Overview of the proposed RAG-LLM framework.

enhance power system operator SA during grid operations. As illustrated in Fig. 1, the system integrates several stages: (i) Operator query input concerning emergency procedure lookup, compliance interpretation, or event response; (ii) Hybrid retrieval combining BM25 sparse matching, FAISS dense vector search, and RRF to identify and rank relevant documents from the local knowledge base, followed by context assembly where the prompt is formulated by concatenating retrieved passages, the operator query, and system instructions; and (iii) Response generation using a 4-bit quantized Llama-3.1-8B-Instruct model, which processes the assembled prompt and produces evidence-grounded answers with citations to retrieved documents. The six important components (A−F) of the proposed RAG-LLM framework (see Fig. 1) in coordination with power system operator queries are described below.

A. Local Knowledge Base

The local knowledge associated with our proposed RAG-LLM framework comprises regulatory documents governing power system reliability and emergency operations across North America. Document acquisition, preprocessing, and indexing occur offline as a one-time operation, establishing the foundation for runtime context retrieval of non-parametric sources. The corpus integrates 20 documents (≈86 MB) for operational situational awareness, and these documents are categorized into following three types. ***First,*** the complete set of mandatory and enforceable NERC standards establishes uniform requirements for all entities responsible for maintaining the reliability of the Bulk Electric System (BES). Critical emergency operation standards such as EOP-004-4 (Event Reporting) define reportable events (e.g., loss of firm load and BES emergencies), while BAL-003-02 (Frequency Response and Frequency Bias Setting) governs frequency control during disturbances. ***Second,*** regional operating procedures from four major ISOs (CAISO, ISO-NE, IESO, PJM) translate high-level reliability standards into actionable protocols. Examples include PJM Manual 13 (Emergency Operations) and Manual 03 (Transmission Operations), which detail emergency response coordination and real-time decision making. Additional federal utility procedures from Western Area Power Authority (WAPA) and Puerto Rico Power Authority (PRPA) extend coverage to municipal and islanded grids. ***Third,*** historical incident reports

covering major grid disturbances, such as the 2003 Northeast Blackout, are included to capture event progression, operator response patterns, and lessons learned.

Documents undergo automated preprocessing to enable efficient retrieval during runtime operations. PDF documents are converted using PyPDF extraction; HTML-formatted sources retain their structure via parsing. The chunking strategy employs a structure-aware approach with paragraph boundary detection and section preservation. Each document is segmented into chunks targeting 3,500 characters with a 400 character overlap (11.4% overlap ratio) and a maximum size of 6,000 characters to preserve section integrity. This overlap preserves contextual boundaries between adjacent segments, ensuring that multi-step procedures or requirements clauses spanning chunk boundaries remain accessible during retrieval. Character-based chunking was selected over token-based methods to decouple the preprocessing stage from model-specific tokenization schemes, enabling flexibility across different embedding models and language models. Each chunk is indexed with metadata preserving source document identifier and sequential chunk position, enabling full citation traceability in operator responses.

The preprocessing pipeline generated 2,773 indexed chunks stored in parquet format for efficient columnar access during retrieval operations. For sparse retrieval, a BM25 index is constructed using the rank-BM25 library. For dense retrieval, chunks are encoded using the sentence-transformers/all-MiniLM-L6-v2 model (384-dimensional embeddings) and indexed in a FAISS IndexFlatIP structure for efficient cosine similarity search via inner product computation.

B. Power System Operator Query Interface

The runtime framework begins with the operator query module where the user formulates a natural-language request related to emergency procedures, compliance interpretation, or event response. These queries typically arise under time-critical operating conditions, requiring grounded and document-based responses. The query is then passed to the retrieval engine for context generation.

C. Hybrid Retrieval

The retriever module integrates sparse and dense retrieval mechanisms to balance lexical precision with semantic recall across diverse query types. The hybrid architecture operates in three stages: candidate generation from dual retrievers, query enhancement, and rank fusion [15]. Before scoring, queries undergo pattern-based expansion to bridge semantic gaps between the natural language query and technical terminology within documents. For example, when queries contain patterns indicating event reporting (e.g., "reportable" and "events"), the system appends domain-specific terms such as "MW" and "transmission." This expansion addresses semantic gaps observed when operators use natural language to query technical specifications containing precise numerical thresholds and regulatory terminology.

Sparse Retrieval with BM25. The BM25 component performs keyword-based matching across all tokenized chunks. To improve precision for regulatory queries, scores are augmented through intelligent boosting: chunks containing structural markers (e.g., "attachment", "section") receive a 2.5× boost for list-based queries, while chunks with "applicability" or "purpose" sections are given a 3.0× boost for regulatory and standard-specific queries. These boost multipliers were calculated heuristically to prioritize precision in high-stakes operator queries. The system retrieves the top-k_{BM25}=30 candidates, with dynamic expansion to k_{BM25}=100 for detected list queries requiring comprehensive coverage.

Dense Retrieval with FAISS. The dense retrieval component employs FAISS with pre-computed sentence embeddings. During preprocessing, each chunk is encoded using sentence-transformers/all-MiniLM-L6-v2 model, producing 384-dimensional embeddings. At query time, the query undergoes identical encoding, and FAISS retrieves the top k_{FAISS}=30 candidate chunks via cosine similarity computed as inner product over normalized vectors. Dense retrieval captures semantic relationships independent of lexical (keyword) overlap, complementing BM25's keyword-based approach for queries where meaning diverges from surface form.

Reciprocal Rank Fusion. The final ranking combines BM25 and FAISS candidate sets using RRF [16]. For each chunk d in either candidate set, RRF score is computed as

$$\text{RRF}(d) = \frac{1}{K + r_{BM25}(d)} + \frac{1}{K + r_{FAISS}(d)} \tag{1}$$

where $r_{BM25}(d)$ and $r_{FAISS}(d)$ denote the rank assigned by each retriever with penalty rank $\max(k_{BM25},\ k_{FAISS}) + 1$ for missing entries, and K=60 is the RRF constant. An RRF constant of K=60 provides balanced smoothing of rank penalties, giving more weight to mid-ranked documents while still favoring top-ranked results. This prevents domination of a single retriever's top-1 ranking from overwhelming the fusion. Chunks are sorted by descending RRF score, and the top k=15 are returned for context assembly, balancing comprehensive coverage with computational efficiency. RRF provides several advantages: (i) it is parameter-free beyond K, (ii) it naturally handles cases where retrievers disagree, and (iii) it mitigates the impact of outlier rankings from individual systems.

D. Context Assembly

Following retrieval, ranked chunks are assembled into a structured prompt designed to enforce citation discipline and reduce hallucination risk. In our proposed framework, the assembled prompt consists of three major components: (i) system instructions that constrain the LLM to answer only with claims grounded by retrieved context, (ii) ranked context chunks with metadata, and (iii) the operator query. The system instructions define the model's role as a power systems operations assistant and impose strict constraints, i.e., the model must answer solely using information from retrieved context, cite the source document/chunk for each claim, and explicitly state when any information is missing from the provided documents. When uncertain or lacking sufficient context, the model is instructed to flag missing information rather than inferring or extrapolating from general knowledge. These strict constraints are placed to limit the model's likelihood of hallucinating during high-stakes operations.

Retrieval chunks are presented to the model with explicit provenance metadata. Each chunk is formatted with a header indicating its source documents and sequential position (e.g., "[DOC BAL-003-02 CHUNK 5]"), followed by the chunk text. Chunks are separated using horizontal delimiters and a new chunk is indicated by a new header. The top three highest-ranked chunks are always included with additional chunks added in descending RRF score order. This ensures critical information is always included as context in the prompt.

E. Large Language Model Processing

Response generation employs a locally deployed Llama 3.1-8B-Instruct model quantized to 4-bit precision. This model processes the assembled prompt using llama.cpp, a high-performance C++ inference engine optimized for low-latency local execution. Quantization reduces the model memory footprint from approximately 16 GB (IEEE FP16 precision) to 2–3 GB (4-bit), enabling deployment on commodity GPU hardware without requiring enterprise-grade accelerators. The 4-bit quantized model maintains inference quality while achieving significant reductions in memory bandwidth requirements; and this is critical for real-time operator decision-making.

The model runs on an NVIDIA RTX 3090 GPU (24 GB VRAM, 10,496 CUDA cores) with inference parameters configured to prioritize factual accuracy and reproducibility over creative generalization. The llama-server is configured with a context window of 16,384 tokens to accommodate large RAG prompts and a maximum output length of 2,048 tokens for comprehensive answers. Inference parameters prioritize factual accuracy and reproducibility over creative responses: temperature is set to 0.0 to produce deterministic answers, and a repeat penalty of 1.05 reduces redundant phrasing. Latency measurements captured during runtime include both retrieval time and LLM inference (generation) time, enabling performance profiling for real-time operational deployment.

F. Evidence Grounded Response

The final stage of the framework returns an evidence-grounded response to the operator, enforcing strict citation discipline and minimizing hallucination risk. Upon receiving the assembled prompt, the Llama 3.1-8B-Instruct model generates a comprehensive answer constrained by system instructions to cite only claims supported by the retrieved context. Each response is post-processed to verify compliance with grounding requirements. The system parses the output to detect citations in the form [DOC: filename, CHUNK: number], ensuring every claim references its source context. For list-type queries (e.g., "Which events are reportable under EOP-004-4?"), the response is post-processed to enumerate items cited from the retrieved context. The system counts the listed items and checks for completeness markers. If the model omits items present in the retrieved chunks, fails to provide required citations, or does not produce a complete listing, the response is flagged with incompleteness and returned to the operator.

If the evidence required to answer the query is not present in the provided context, the model is instructed to explicitly state *"This information is not covered in the provided context."* This guarantees that unsupported or hallucinated answers are not presented in high-stakes operational scenarios.

III. Experimental Setup and Evaluation Metrics

A. Evaluation Dataset

For quantitative evaluation, a domain-specific evaluation set comprising natural-language operator queries, ground-truth instructor answers, generated LLM responses, and supporting chunks for each query was constructed. The dataset contains N=30 samples arranged across the three dimensions of operational SA: 10 queries targeting emergency procedure lookup (perception), 10 for targeting compliance interpretation (comprehension), and 10 for event response guidance (projection). This stratification ensures balanced coverage and enables quantitative analysis of system performance for each SA dimension. This paper employed the Retrieval-Augmented Generation Assessment Suite (RAGAS) framework to evaluate four metrics using GPT-3.5-turbo as an LLM judge [17].

B. Automated Metrics: RAGAS Evaluation

Factual consistency and groundedness of generated responses were assessed using the RAGAS evaluation framework, leveraging GPT-3.5-turbo as a judging LLM. Four metrics are computed for each question-answer pair. Each evaluation instance used the top-k=15 retrieved chunks from the hybrid retrieval module, consistent with system runtime configuration. For retrieval evaluation, four metrics are described below.

Faithfulness: It measures the proportion of model-generated claims that are factually supported by retrieved context. The generated answer is decomposed into atomic statements, and each statement is verified against retrieved contexts for factual accuracy using the LLM judgment. For a set of statements S,

$$F = \frac{|\{s \in S \mid s \text{ is supported by context}\}|}{|S|} \tag{2}$$

where F denotes the faithfulness score, s represents an individual generated statement, and $|S|$ the cardinality of S.

Answer Relevancy: It measures how well the generated answer addresses the original question through reverse question generation. GPT-3.5-turbo generates N=3 artificial questions from the response, and relevancy is computed as the mean cosine similarity between the embeddings of the original question and the generated questions, defined as

$$A_r = \frac{1}{N} \sum_{i=1}^{N} \frac{E(q) \cdot E(g_i)}{|E(q)| \cdot |E(g_i)|} \tag{3}$$

where A_r is the answer relevancy score, q is the original question, g_i is the i-th generated question, $E(\cdot)$ denotes the embedding function, "$\cdot$" represents the dot product between embedding vectors, and $|E(\cdot)|$ is the Euclidean vector norm.

Context Precision: It evaluates whether LLM-judged useful contexts are ranked higher in retrieval results. For each retrieved context at position k (ranging from 1 to 15), GPT-3.5-turbo determines its relevance $v_k \in \{0, 1\}$ by assessing its utility for answering the question given the ground truth. The metric rewards retrieval systems that place useful contexts earlier in the ranked list, and it is defined as

$$P_c = \frac{\sum_{k=1}^{K} (\text{Precision@}k \times v_k)}{\sum_{k=1}^{K} v_k} \tag{4}$$

where P_c denotes the context precision score and Precision@$k = \frac{\text{\# relevant contexts in top-}k}{k}$ measures the proportion of relevant contexts up to position k.

Context Recall: It measures whether the retrieved contexts contain all information needed to answer the query. The ground truth is decomposed into statements, and each statement is checked for attribution to at least one retrieved context. For ground truth statements G, context recall is defined as

$$R_c = \frac{|\{g \in G \mid g \text{ can be attributed to retrieved contexts}\}|}{|G|} \quad (5)$$

where R_c is the context recall score, g represents an individual ground-truth statement, and $|G|$ denotes its cardinality.

C. Retrieval Quality Metrics

To assess whether the retrieval system provides context chunks that are actually utilized by the LLM, we introduced citation-based variants of Precision@K and Recall@K. Traditionally both require exhaustive relevance annotations across the entire corpus for each query, which is impractical for a large corpus such as the one we utilized, i.e., ≈86 MB. Instead, we leveraged the LLM-generated citations as an operational proxy for relevance: if a chunk is cited in a response, it contributes meaningfully to the response. For a given query, let $C^K_{\text{retrieved}}$ represent set of top-K retrieved context chunks, and let C_{cited} denote the set of chunks explicitly cited in the LLM-generated answer. These citation-based variants are described below.

Citation Precision@K: It quantifies the fraction of the top-K retrieved chunks that are explicitly cited in the generated answer, thus measuring retrieval efficiency. Formally, $P_{\text{cite}}@K$ is defined as

$$P_{\text{cite}}@K = \frac{|C^K_{\text{retrieved}} \cap C_{\text{cited}}|}{K} \quad (6)$$

Citation Recall@K: It measures the fraction of context chunks cited in the answer that appear within the top-K retrieved chunks, evaluating retrieval coverage. $R_{\text{cite}}@K$ is defined as

$$R_{\text{cite}}@K = \frac{|C^K_{\text{retrieved}} \cap C_{\text{cited}}|}{|C_{\text{cited}}|} \quad (7)$$

where $|C|$ represents the cardinality of set C. Results are averaged over all evaluation queries and reported for $K \in \{3, 5, 10, 15\}$ to quantify how well the retrieval system supports the LLM's grounded answer generation.

IV. Experimental Results and Discussion

This section presents the quantitative and qualitative evaluation of the proposed RAG-LLM framework. Evaluation results are organized into two parts: (i) automated grounding metrics computed via the RAGAS framework and (ii) retrieval quality metrics based on citation alignment between the retrieved contexts and those explicitly cited by the model within the response. All evaluations are performed on the $N = 30$ domain-specific query dataset described in Section III, encompassing the three SA dimensions: (i) perception, (ii) comprehension, and (iii) projection.

Table I summarizes the RAGAS evaluation results across the three query types corresponding to the three SA dimensions.

TABLE I
RAGAS Evaluation Scores by SA Dimension

Metric	Procedure Lookup	Compliance Interpretation	Event Response	Overall
Faithfulness	0.7921	0.7357	0.971	0.831
Answer Relevancy	0.9859	0.9697	0.9584	0.9722
Context Precision	0.9393	0.9468	0.9065	0.9257
Context Recall	0.925	0.96	0.975	0.9548

TABLE II
Citation-Based Retrieval Performance by Chunk

K	Citation Precision@K	Citation Recall@K
3	0.2000	0.2589
5	0.2067	0.4006
10	0.2033	0.8033
15	0.1489	0.8767

Overall, the proposed RAG-LLM system achieved strong factual grounding and semantic alignment, reflected by high average faithfulness and answer relevancy scores across all query types. The average faithfulness across all queries was 0.831, indicating high factual groundedness within the LLM-generated response, though this metric yielded the lowest relative score among the four. Lower faithfulness scores were observed in the procedure lookup (0.7921) and compliance interpretation (0.7357) categories, suggesting occasional generation of statements not explicitly grounded in retrieved context. In contrast, event response queries achieved a faithfulness score of 0.971, outlining the model's ability to provide grounded answers in high-stakes scenarios. Answer relevancy was particularly high, exceeding 0.95 across all query types, indicating the LLM-generated responses remained well-aligned with operator query intent, confirming the model's ability to maintain semantic coherence even when synthesizing multi-document contexts.

The system demonstrates strong retrieval quality across all categories, with context recall exceeding 0.92 and context precision exceeding 0.90. Context recall above 0.92 indicates that the hybrid retrieval approach successfully captured most of the information necessary to answer the ground truth queries. Context precision above 0.90 demonstrates useful contexts were consistently ranked in top positions, validating the efficiency of the proposed hybrid retrieval framework.

Among all categories, the proposed system performed exceedingly well in event response, with values exceeding 0.90 for all metrics. These results suggest the proposed RAG-LLM framework effectively delivers highly reliable and contextually relevant answers in operational environments. Performance in procedure lookup and compliance interpretation that tend to involve more technical and precise requirements was found to be slightly lower, and this is likely due to the complexity and specificity of these categories.

As we can observe in Table II, Citation Precision@K values remain relatively stable around 0.20 across all K values with a slight drop (0.1489) at K=15. This implies that approximately 20% of the top-K (i.e., K = 3, 5, and 10) retrieved chunks were explicitly cited in the LLM-generated answers regardless of the number of candidates present. Furthermore, the drop at K=15 indicates that more non-cited chunks are included as K

increases, diluting the fraction of cited passages.

Citation Recall@K, which measures retrieval coverage, is found to be increasing significantly with a larger K (see Table II), rising from about 0.26 at K=3 to 0.88 at K=15. This clearly suggests that nearly all cited context chunks are returned by the retriever when up to 15 candidates are considered, which highlights the strong recall performance. Hence, expanding the candidate set increases the likelihood that all chunks cited by the LLM are available in retrieval results.

From the obtained experimental results, we can further observe that the modest Precision@K values reflect the broad and complex nature of operator queries and supporting documents. Moreover, while the system is highly effective at recalling most of the cited contexts, only a subset of candidates are ultimately cited in the LLM-generated response. This is typical in RAG frameworks where ensuring complete recall is prioritized, even if irrelevant or redundant passages are included within the greater retrieved context [18]. While Recall@15 approached 0.88, it did not reach 1.0 because of several model-generated citations referenced document chunks that were not among the top-15 retrieved contexts. This is primarily due to plausible yet nonexistent references created by the model, i.e., citations to documents it had not seen in the prompt.

As demonstrated by the test results, this paper significantly contributed to improving decision support as well as situational awareness for power system operators, in particular by developing a hybrid RAG system and LLM with domain-specific sources to support critical queries. Evaluation results thus underscore the importance of effective retrieval and grounded generation.

V. Conclusion

This paper presented a novel retrieval-augmented large language model (RAG-LLM) framework for enhancing power system operator situational awareness across three operational query types: procedure lookup (perception), compliance interpretation (comprehension), and event response (projection); each type is mapped to a dimension of situational awareness. The hybrid retrieval mechanism combining BM25, FAISS, and RRF enabled efficient multi-source document retrieval with strong grounding performance. Quantitative evaluation using RAGAS demonstrated high factuality and semantic alignment, with average faithfulness of 0.83, answer relevancy of 0.9722, context precision of 0.9257, and context recall of 0.9548. Test results clearly outlined the effectiveness of hybrid retrieval-augmented reasoning for decision support in high-stakes power system operations. Citation-based analysis further demonstrated strong coverage across retrieved contexts, with Recall@K increasing from 0.26 to 0.88 for K=3 and 15, respectively, indicating nearly all cited evidence was retrieved by the hybrid system. Precision@K remained consistent around 0.20 across all values of K, reflecting the broad nature of operator queries where only a subset of retrieved chunks contribute to the response. These findings demonstrated that the system achieves high recall without sacrificing grounding reliability, which is an essential characteristic of power systems operator support. Collectively, the proposed RAG-LLM framework contributed significantly to advancing situational awareness within power system operations as well as establishing a solid foundation for continued research aimed at modernizing and strengthening the electric grid evolution.

Future work could be interesting to focus on expanding the RAG-LLM framework to support multi-hop reasoning across cross-referenced reliability standards and dynamic retrieval adaptation during evolving events and prompts. Incorporating temporal and event-state awareness would further allow the system to track progression of disturbances in real-time, thereby updating context as new telemetry and operator reports become available.

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